Looking for a frequency-based principle to predict the sensitivity of convolutional neural networks to Gaussian image perturbation

Work-in-progress

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Background and motivation

Acquisition conditions heterogeneity dramatically affect the accuracy of remote PV detectors [2].
Image corruptions [1], especially Gaussian blur, conveniently model image acquisition conditions.

Our contribution

Contribute to **understanding how image acquisition conditions affects CNNs** to build more trustworthy systems.

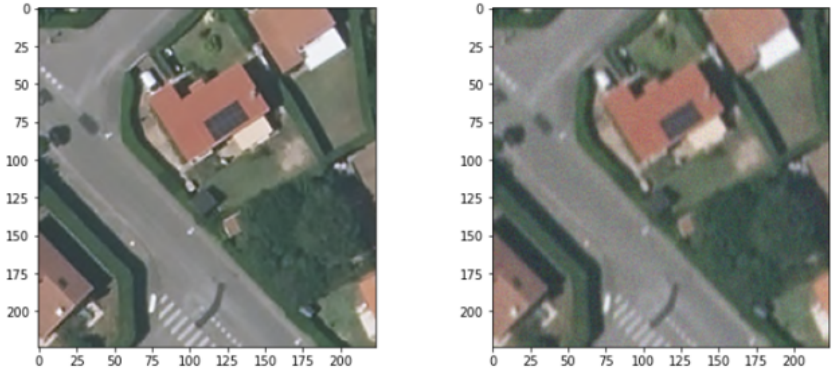
Consistency of predictions before and after blurring

Binary classification task of solar panels from the dataset of [3]. Apply a random **Gaussian blur** g_σ, $\sigma \in [0.1, 1]$ on the images X and replicate the inference:

$$\tilde{X} = g_\sigma * X$$

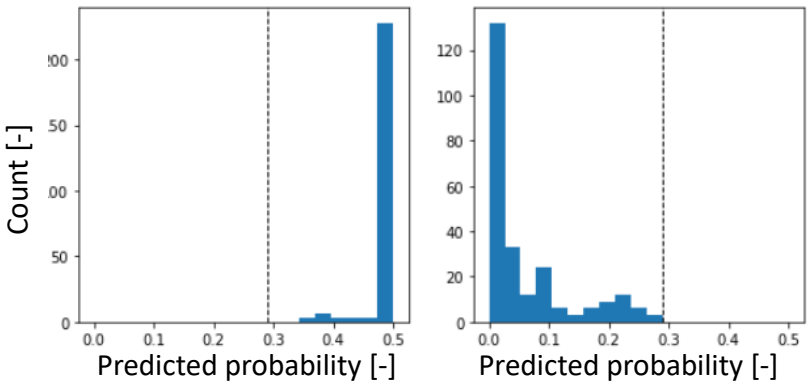
Measure the **consistency** [6] of the predictions by the model f between the source and altered images:

$$\mathbb{E}\left(1\left\{f(\tilde{X}) = f(X)\right\}\right)$$



Stylized facts

1. Images are unevenly affected by the blurring (consistency of 98% in our framework)
2. The predicted probability shifts: the model is confident about its prediction, <u>but reversed</u> : *the blurring removed the frequencies needed for prediction*



How can we explain this result?

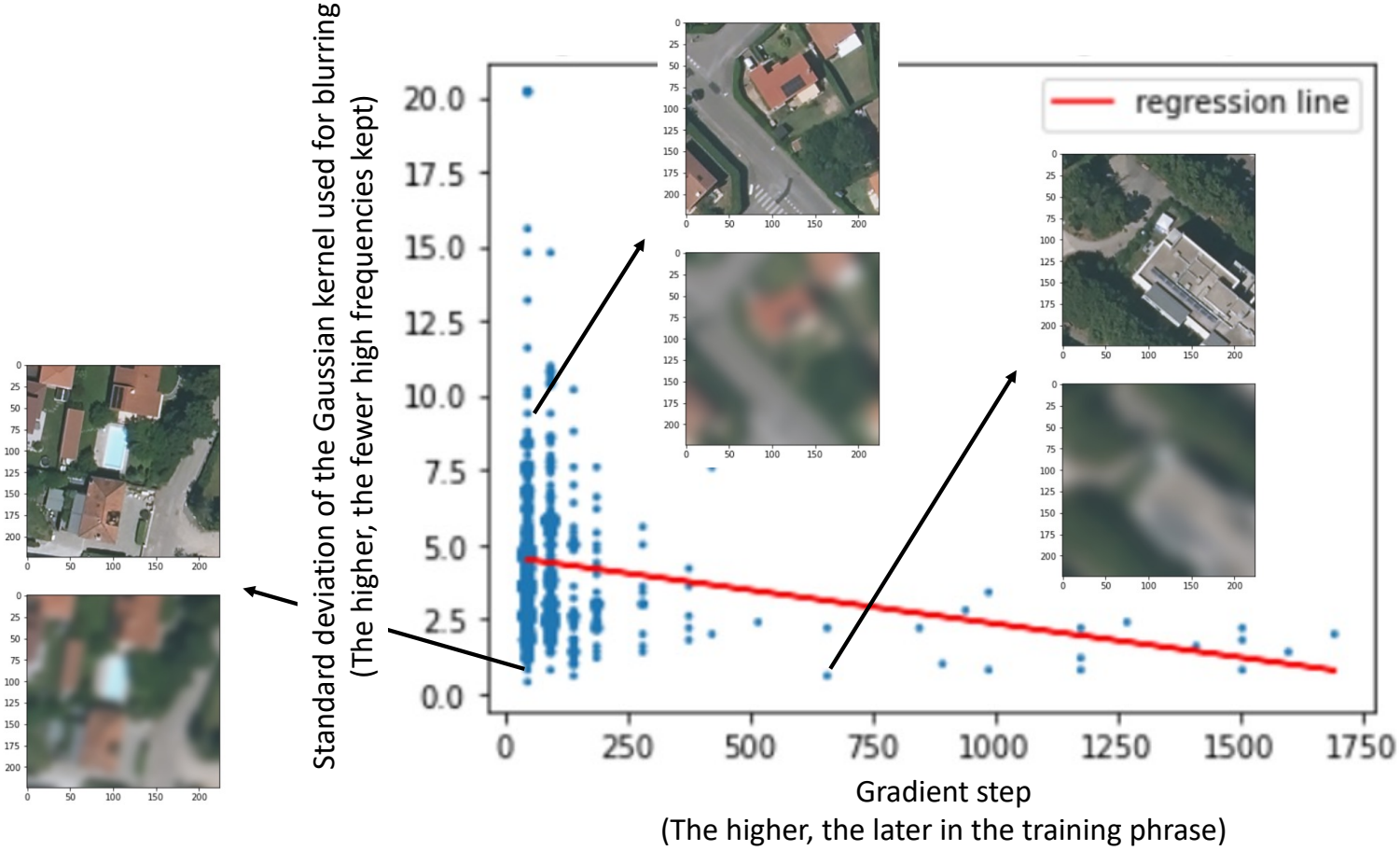
Frequency principle [4, 5]

Deep neural networks fit the function f that maps the inputs X to the outputs y gradually, from its lowest frequencies to its highest.

A corollary is that the model correlates input features *with gradually increasing frequencies* to the labels during learning.

Hypothesis: The value of the "pivotal frequencies", i.e. the lowest frequency needed to *flip* the prediction, depends on the *moment* it is learned. The sooner the lower.

Results partially align with the theory



Conclusions and future work

Conclusions:

1. All images are not affected by the blurring.
2. The predicted probability shifts: the model is confident about its prediction, *but reversed*.
3. The *moment* an image is learned is only a partial predictor for its "pivotal frequencies"

Future works:

1. Quantify the "pivotal frequencies" ;
2. Understand how a model picks frequencies during learning ;
3. Generalize to benchmark datasets (e.g., [1]) and other corruptions

Thank you for your attention!

Questions?

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Unveiling *pivotal frequencies* in binary classification

- CNN: Function f that correlates an input X to a label $y \in \{0, 1\}$
- The frequencies ω composing X can be isolated (FFT, ...)

The **pivotal frequency** ω^* is the frequency *needed for the prediction*. If the perturbation $\tilde{X}$ of X does not contain ω^*, then f *flips* its prediction for X

- Robustness: how *transferable* (across datasets) the pivotal frequency is.
- We need **characterize how a model converges towards the pivotal frequency**